The Promised Trillion Dollar AI-Coding Market

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> Who projects the AI coding market, what they project, and what every projection assumes: one path to autonomous computer automation, with high energy, high compute, per-token rent and the model in the loop forever. The Reality Code Engine (RCE) measures a second path. With correctness as the gate and joules as the ranking, it resolved 20 of 20 repair tasks with 0 model calls. Where that path covers a task, the marginal cost falls toward the energy to run it, and the projections lose their base.

The economics companion to The Reality Code Engine: One Form, Many Fits, and the Low-Entropy Path to Autonomous Coding. The research paper holds the engineering, the theory and the measurements. This essay is where the theory and the economics meet.

Abstract

The trillion-dollar AI coding market is David Charlot's informed hyperbolic claim, and the real projections bear out its scale. A venture estimate puts the value of the world's software developers at about $3 trillion a year and calls AI coding a market that can support "even a trillion-dollar giant" [1]. Two AI coding companies reported over 1 billion and over $2.5 billion in annualized revenue within a year of each other [5, 6]. One frontier lab is valued at 380 billion, and its coding product is its showcase [6]. One provider's token volume grew from 9.7 trillion to over 3.2 quadrillion tokens a month in two years [8]. Capital expenditure by the largest technology companies exceeded $400 billion in 2025 [10].

Every one of these projections assumes the same thing. There is only one path to autonomous computer automation. It is high energy, high compute, high entropy and disorder. A model sits in the loop for every task, and every task pays rent per token.

The RCE research paper measures a second path. A program is a database functional state machine: one Form, many fits. Most fits already exist. Correctness is the gate, and joules are the ranking. On a frozen 20-task repair corpus, RCE resolved 20 of 20 with 0 model calls (RCE §6.1). Where that path covers a task, the marginal cost of the task falls toward the energy to run it, and a market priced on generation loses its base on that task. This essay separates the projections from the measurements, names what each projection assumes, and states where it breaks.

Shared thesis, stated once for every paper. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. The gate form defines what that means. Among runs that pass the task's completeness predicate $C(z)$ within the stated budget and deadline,

systems are ranked by joules to completion, J^*. Correctness, deadlines and task size are the gate, not terms in the score. Parameters, tokens, benchmark scores and prices carry zero weight in the ranking (Agency, Section 1). Estimates are never `measured_j`.

Price, a separate claim, for the economics companions. A price trend is a separate claim, labeled as one. Epoch AI reports that the cost of a given level of AI performance has fallen about 47 percent per quarter since 2023, about 13 times per year (Emberson and Roodman, *The plunging price of thought*, 22 September 2026). If that trend continues, energy becomes the dominant marginal cost of inference. That is a cited trend with a condition, not a measured collapse. Spell Check is Global is an existence proof that some computer intelligence commoditizes to bundled zero price. The theory these companions use is in the agency paper, Metered Commit Gates and The Reality Code Engine.

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1. Projections and measurements are different objects

Every number below carries one of five labels.

Label	Meaning
Projection	A forecast or a value estimate built on stated assumptions
Company-reported	A figure a company states about itself, such as annualized run-rate revenue (the latest month times twelve) or a funding valuation
Survey estimate	A market-sizing figure from a survey of buyers
Filed	A figure in a regulatory filing
Measured	A count or an energy reading from a run whose artifacts are kept. In RCE, energy carries its own label: `est_j`, `reported_j` or `measured_j`

Only the last row is evidence about the cost of a correct result. The others are evidence about what the market expects.

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2. Who projects what

2.1 Developer value: the trillion

Claim	Number	Label	Source
Software developers worldwide	about 30 million (27 million to 47 million by estimate)	Projection input	Appenzeller and Li [1]
Value per developer per year	$100,000 (assumed)	Projection input	[1]
Annual economic contribution	about $3 trillion	Projection	[1]

Claim	Number	Label	Source
Productivity gain from a simple assistant	about 20%	Estimate from interviews	[1]
Productivity gain from a best-of-breed deployment	“at least double”	Estimate “based on anecdotal evidence”	[1]

The same essay says the AI coding ecosystem can “support dozens of billion-dollar companies and even a trillion-dollar giant” [1]. That sentence is the trillion. It is a value estimate. It is not a revenue forecast.

2.2 Tool revenue and enterprise spend

Claim	Number	Label	Source
Enterprise spend on AI coding tools, 2025	$4.0 billion, 55% of departmental AI spend	Survey estimate	Menlo Ventures [2]
Enterprise generative AI spend, 2025	$37 billion	Survey estimate	[2]
AI code tools market, 2026	$9.35 billion, to 29.96 billion by 2031	Projection	Mordor Intelligence [3]

The revenue the market reports today is in the billions. The trillion is the value the market hopes to capture.

2.3 Company revenue and valuations

Company claim	Number	Date	Label	Source
Cursor annualized revenue	over $1 billion	13 November 2025	Company-reported	Cursor [5]
Cursor Series D	$2.3 billion at 29.3 billion post-money	13 November 2025	Company-reported	[5]
Anthropic run-rate revenue	$14 billion	12 February 2026	Company-reported	Anthropic [6]
Claude Code run-rate revenue	over $2.5 billion	12 February 2026	Company-reported	[6]
Anthropic Series G	$30 billion at 380 billion post-money	12 February 2026	Company-reported	[6]

Anthropic states that the round funds “infrastructure expansions” [6]. Revenue, valuation and compute are one bet.

2.4 Inference, tokens and unit economics

Claim	Number	Label	Source
Google tokens processed per month	9.7 trillion (2024), about 480 trillion (2025), over 3.2 quadrillion (May 2026)	Company-reported	Pichai [8]
Google internal AI developer tools	half a trillion tokens a day in March 2026, more than three trillion a day in May	Company-reported	[8]
Cost of one agentic coding query at list prices	$2.50 : a 100,000 - token context and 10,000 output tokens at 15 and $75 per million tokens	Illustration	Appenzeller and Li [1]
The same at 3 queries an hour, 7 hours a day, 200 days	about $10,000 a year per developer	Illustration	[1]
Tokens in agentic coding against code reasoning and code chat	about 1,000 times more; runs of one task differ by up to 30 times; higher token use "does not translate into higher accuracy"	Measured (eight models, SWE-bench Verified)	Bai, Huang and Wang [9]
Net losses from Microsoft's investment in OpenAI, quarter ended 30 September 2025	$4.1 billion, primarily its equity-method share	Filed	Microsoft Form 10-Q [7]

Two lines in the sources state the economics plainly. Pichai: "many companies are already blowing through their annual token budgets, and it's only May" [8]. Appenzeller and Li: "For decades, software development cost was almost purely people's cost, but now LLMs add a substantial opex component" [1].

2.5 Price per token

Claim	Number	Label	Source
Cost of a fixed level of AI performance	down about 47% per quarter, about 13 times per year, since 2023	Measured price series (five benchmarks)	Epoch AI [4]

2.6 Energy and capital

Claim	Number	Label	Source
Data-centre electricity, 2024	about 415 TWh, about 1.5% of world electricity	Estimate	IEA, *Energy and AI* [11]
Data-centre electricity, 2030	about 945 TWh, Base Case	Projection	[11]

Claim	Number	Label	Source
Updated: 2025 to 2030	485 TWh to 950 TWh, about 3% of world demand	Estimate and projection	IEA, *Key Questions on Energy and AI* [10]
Capex of the largest technology companies	over $400 billion in 2025, expected up another 75% in 2026	Reported and projected	[10]
Alphabet capex	$31 billion in 2022; about 180 to $190 billion expected in 2026	Company-reported and guidance	Pichai [8]
Energy per task	down by at least an order of magnitude a year; reasoning and agentic tasks use "hundreds or thousands of times more energy per query than simple text generation"	Estimate	[10]
One median text prompt to a production assistant	0.24 Wh	Measured by the operator	Elsworth et al. [12]

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3. What the projections assume

Read together, the projections rest on four assumptions. Each is a statement about the path, not about the value of software.

1. **One path: generation.** Every change is written by a model. The value estimate scales productivity by the model, and the cost illustration prices every query [1].
2. **High energy and high compute.** Token volume, data-centre electricity and capex are projected to grow together [8, 10, 11]. The energy per query falls, and the tokens per task rise faster [9, 10].
3. **Per-token rent.** Revenue is billed per token or per seat whose cost is tokens. A developer's annual token cost is projected at the scale of a salary [1].
4. **AI in the loop forever.** Task 10,000 costs what task 1 costs, because nothing the model learned on task 1 runs without the model on task 10,000.

The essay that states the trillion also states its limit: "while substitution pricing works in early markets, eventually the cost of a good converges to its marginal cost, and so does pricing" [1]. This essay agrees, and asks what the marginal cost of a correct coding task is.

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4. The theory that prices the task

The RCE research paper supplies three results that turn the question into a quantity.

Correctness is the gate. A task is done when its completeness predicate $C(z)$ holds: its tests pass, a proof exists, or an agreement trace matches (RCE §1). A buyer pays for $C(z) = 1$, not for tokens.

Joules are the ranking. Among systems that pass the gate, the one with the fewest joules to completion, J^*, wins (Agency §1). In a competitive market, sellers that pass the same gate compete on cost, and the cost floor of a task is the energy to complete it plus the price of the proof.

The Form makes most fits reusable. A program is a tuple $P = (S, E, \delta, D, \lambda)$, and by Gurevich's Abstract State Machine theorem every sequential algorithm has that shape (RCE §3). Programs differ in fit, and most fits exist already: "70% of the code on GitHub consists of clones of previously created files" (RCE §7). Equivalent fits pass the same gate, so extraction picks the least-joule one (RCE Proposition 3).

Proposition E0 (price of a covered task). Let a task be covered when a stored fit passes its gate. Among sellers that pass the gate, competition drives the price of a covered task toward $c_J \cdot J^*_{\text{floor}} + c_{\text{proof}}$, where c_J is the price of a joule, J^*_{floor} is the energy to retrieve, apply and check the fit, and c_{proof} is the cost of the check. Generation enters only on uncovered tasks.

Mixture of Limits is the philosophy behind the proposition: there is a minimal amount of anything to accomplish a task, and any more is waste (Metered Commit Gates, Part II). Per-token rent past the floor bills for waste.

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5. The measurements

These are measurements, labeled as RCE labels them. They come from the RCE research paper.

Measurement	Result	Label	Where
Frozen 20-task repair corpus	20 of 20 resolved, 0 model calls; 18 fixes byte-identical to the human fix, 2 equivalent	Measured counts	RCE §6.1
The same corpus, earlier solver	10 of 20, 3.10 model calls per re-solved task	Measured counts	RCE §6.1
Energy per solved task, new solver	84.2 J, 20 of 20, 154 s	`reported_j` (RAPL package counter less a stated idle baseline)	RCE §6.2
Energy per solved task, earlier solver	712.4 J, 11 of 20, 29 model calls, 695 s	`reported_j`	RCE §6.2
Energy to apply the known fix and test it (gold)	2.0 J per task	`reported_j`	RCE §6.2

Measurement	Result	Label	Where
First soak with a production model resolver	10 model calls in round 1, then 0 in rounds 2 to 5; 50 if the model ran every round	Measured counts	RCE §6.4
Soaks on live traffic	all 12 runs reach a round with zero new escalations before round 20	Measured counts; energy `est_j`	RCE §6.3

`measured_j` is reserved for a wall meter and is empty in every row.

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6. Where the projections break

Per-token rent breaks on covered tasks. In the frozen corpus, a covered repair bought 0 tokens. Between the earlier and the new solver, model calls per resolved task went from 3.10 to 0 and reported joules per solved task fell 8.5 times, from 712.4 to 84.2 [RCE §6.1, §6.2]. A revenue line priced per token earns nothing on a task that needs no token.

AI in the loop forever breaks at promotion. In the first soak the model was called 10 times in round 1 and never again for the same cases [RCE §6.4]. The model compiled itself out of the loop. Task 10,000 does not cost what task 1 cost.

High energy is not required for a correct result. The gate passed at 84.2 J per task with no model, and the floor of the same corpus is 2.0 J [RCE §6.2]. One median text prompt to a production assistant is measured at 0.24 Wh, or 864 J [12]. An agentic coding task uses many prompts, and the IEA puts agentic queries at hundreds or thousands of times the energy of a simple one [10]. These are different tasks on different machines, so they are not a ratio. They are an order of magnitude to test, under Protocol E1.

Two declines multiply. The price per call falls about 13 times a year at fixed performance [4]. Promotion drives calls per task to zero on covered cases [RCE §6.4]. The cost per task is the product of the two. A projection that extrapolates token volume prices only the first.

The Spell Check pattern. Spell Check is Global records a capability whose marginal cost fell to about zero and that complemented a tool everyone already had. It was bundled into the editor and its separate price disappeared, as bundling economics predicts [13, 14]. A covered coding task has the same two properties: its marginal cost falls to the handler floor, and it complements editors and repositories people already have. The prediction is the same: the covered task is bundled, and what stays priced is the uncovered case and the proof.

What survives. The value estimate of section 2.1 can hold. Developers can produce more. What breaks is the claim that the value is captured as rent on generation. Value created at a falling marginal cost is captured by whoever holds the gate, the proof and the low-entropy floor, and much of it goes to the buyer.

7. Demand

Cheaper tasks can raise total demand. The venture essay expects it: “we believe Jevon’s Paradox in this case holds” [1]. Task Satiation vs. Aggregate Rebound in AI Demand separates the two effects. Per-task appetite stops at the completion predicate. Aggregate use can still grow, because cheaper synthesis opens new tasks. The second path changes the first effect and leaves the second open. A larger market of covered tasks is a market whose marginal cost is the floor, not the frontier.

8. Propositions and protocols

Proposition E1 (the cost curves cross). For a stream of tasks in one repository, the cumulative cost per resolved task of the second path falls below that of a generation agent after a finite number of tasks. *Protocol:* run RCE Protocol 8.3 and a plain generation agent on the same issue stream under a wall meter. Report `measured_j` and dollars per resolved task by task index, and the crossing index.

Proposition E2 (price follows the floor). Where a coding capability is covered by retrieval and rules, its market price falls toward the energy to run it plus the cost of the proof. *Protocol:* track list prices of a covered capability class against its measured run energy over time, in the manner of the Spell Check price series.

Proposition E3 (compute scale is not a moat on covered tasks). On tasks the second path covers, the resolve rate does not rise with the compute of the last gear. *Protocol:* run RCE Protocol 8.2 with no model, a small local model and a frontier model as the last gear. Compare resolve rates on the tasks closed before the last gear.

Proposition E4 (the token share of revenue). In a repository where coverage grows, the share of task cost paid as tokens falls with coverage. *Protocol:* log tokens, joules and dollars per closed task by round in the RCE soak with a priced model resolver, labeled, and report the token share of cost by round.

References

[1] G. Appenzeller and Y. Li, “The Trillion Dollar AI Software Development Stack,” 9 October 2025. https://a16z.com/the-trillion-dollar-ai-software-development-stack/

[2] Menlo Ventures, “2025: The State of Generative AI in the Enterprise,” December 2025. https://menlovc.com/perspective/2025-the-state-of-generative-ai-in-the-enterprise/

[3] Mordor Intelligence, “AI Code Tools Market Size, Share and 2031 Growth Trends Report,” fetched 11 October 2026. https://www.mordorintelligence.com/industry-reports/artificial-intelligence-code-tools-market

[4] B. Emberson and D. Roodman, “The plunging price of thought,” Epoch AI, 22 September 2026. https://epoch.ai/publications/the-plunging-price-of-thought

[5] Cursor, “Past, Present, and Future” (Series D announcement), 13 November 2025. https://cursor.com/blog/series-d

[6] Anthropic, “Anthropic raises $30 billion in Series G funding at $380 billion post-money valuation,” 12 February 2026. https://www.anthropic.com/news/anthropic-raises-30-billion-series-g-funding-380-billion-post-money-valuation

[7] Microsoft Corporation, Form 10-Q for the quarter ended 30 September 2025, Note 3 and Item 2. https://www.sec.gov/Archives/edgar/data/789019/000119312525256321/msft-20250930.htm

[8] S. Pichai, “I/O 2026: Welcome to the agentic Gemini era,” 19 May 2026. https://blog.google/innovation-and-ai/sundar-pichai-io-2026/

[9] L. Bai, Z. Huang and X. Wang, “How Do AI Agents Spend Your Money? Analyzing and Predicting Token Consumption in Agentic Coding Tasks,” 2026. arXiv:2604.22750. https://arxiv.org/abs/2604.22750

[10] International Energy Agency, *Key Questions on Energy and AI*, executive summary, 2026. https://www.iea.org/reports/key-questions-on-energy-and-ai/executive-summary

[11] International Energy Agency, *Energy and AI*, executive summary, April 2025. https://www.iea.org/reports/energy-and-ai/executive-summary

[12] C. Elsworth et al., “Measuring the environmental impact of delivering AI at Google Scale,” 2025. arXiv:2508.15734. https://arxiv.org/abs/2508.15734

[13] Y. Bakos and E. Brynjolfsson, “Bundling Information Goods: Pricing, Profits, and Efficiency,” *Management Science* 45(12), 1999. doi:10.1287/mnsc.45.12.1613

[14] C. Shapiro and H. R. Varian, *Information Rules: A Strategic Guide to the Network Economy*, Harvard Business School Press, 1998.